# Worksheet 17 · Simple & Compound Int. IGCSE

*Edexcel IGCSE Mathematics · past paper questions*

| Name: | Subject: Math | Date: |
|---|---|---|
| **Learning Objectives:**<br>Students will be able to:<br>• Calculate simple interest<br>• Calculate compound interest and depreciation<br>• Use multipliers for repeated percentage change | | |
| **Grade:** / 27 | **Teacher's Feedback:** | |

*Answer in the space given. Show your working. Marks for each part are shown in brackets.*

## Key notes

Formula → New = Original × Multiplier$^{\text{years}}$

## Questions *Simple & Compound Int.*

### Question 1 (6 marks)

Jenny invests $8500 for 3 years in a savings account.
She gets 2.3% per year compound interest.

(a) How much money will Jenny have in her savings account at the end of 3 years?
Give your answer correct to the nearest dollar.

$.....
(3)

Rami bought a house on 1st January 2015

In 2015, the house increased in value by 15%
In 2016, the house decreased in value by 8%

On 1st January 2017, the value of the house was $687 700

(b) What was the value of the house on 1st January 2015?

$.....
(3)

### Question 2 (3 marks)

On 1st January 2016 Li bought a boat for $170 000
The value of the boat depreciates by 8% per year.

Work out the value of the boat on 1st January 2019
Give your answer correct to the nearest dollar.

$

### Question 3 (3 marks)

Max invests $6000 in a savings account for 3 years.
The account pays compound interest at a rate of 1.5% per year for the first 2 years.

The compound interest rate changes for the third year.
At the end of 3 years, there is a total of $6311.16 in the account.

Work out the compound interest rate for the third year.
Give your answer correct to 1 decimal place.

..... %

### Question 4 (3 marks)

Jan invests $8000 in a savings account.
The account pays compound interest at a rate of $x$% per year.

At the end of 6 years, there is a total of $8877.62 in the account.

Work out the value of $x$.
Give your answer correct to 2 decimal places.

$x =$ .....

### Question 5 (3 marks)

Shane invests 7200 dollars for 3 years in a savings account.
He gets 2.5% per year compound interest.

How much money will Shane have in his savings account at the end of 3 years?
Give your answer to the nearest dollar.

..... dollars

## Question 6 (6 marks)

Tomas invests £5000 in a savings account for 4 years. The account pays compound interest as follows:

- 4% for the first year,
- 3% for each of the subsequent years.

(a) Work out the value of Tomas's investment at the end of 4 years to the nearest pound.

£.....
**(3)**

Tomas wanted to have at least £6000 in the account at the end of the 4 years.

(b) Work out the least amount of pounds Tomas should have invested to achieve his goal.

£.....
**(3)**
**(Total 6 marks)**

## Question 7 (3 marks)

Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

| Bank **G** | Bank **H** |
|---|---|
| 1.6% per year compound interest | 2.9% interest added after two years |

The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

How much more?